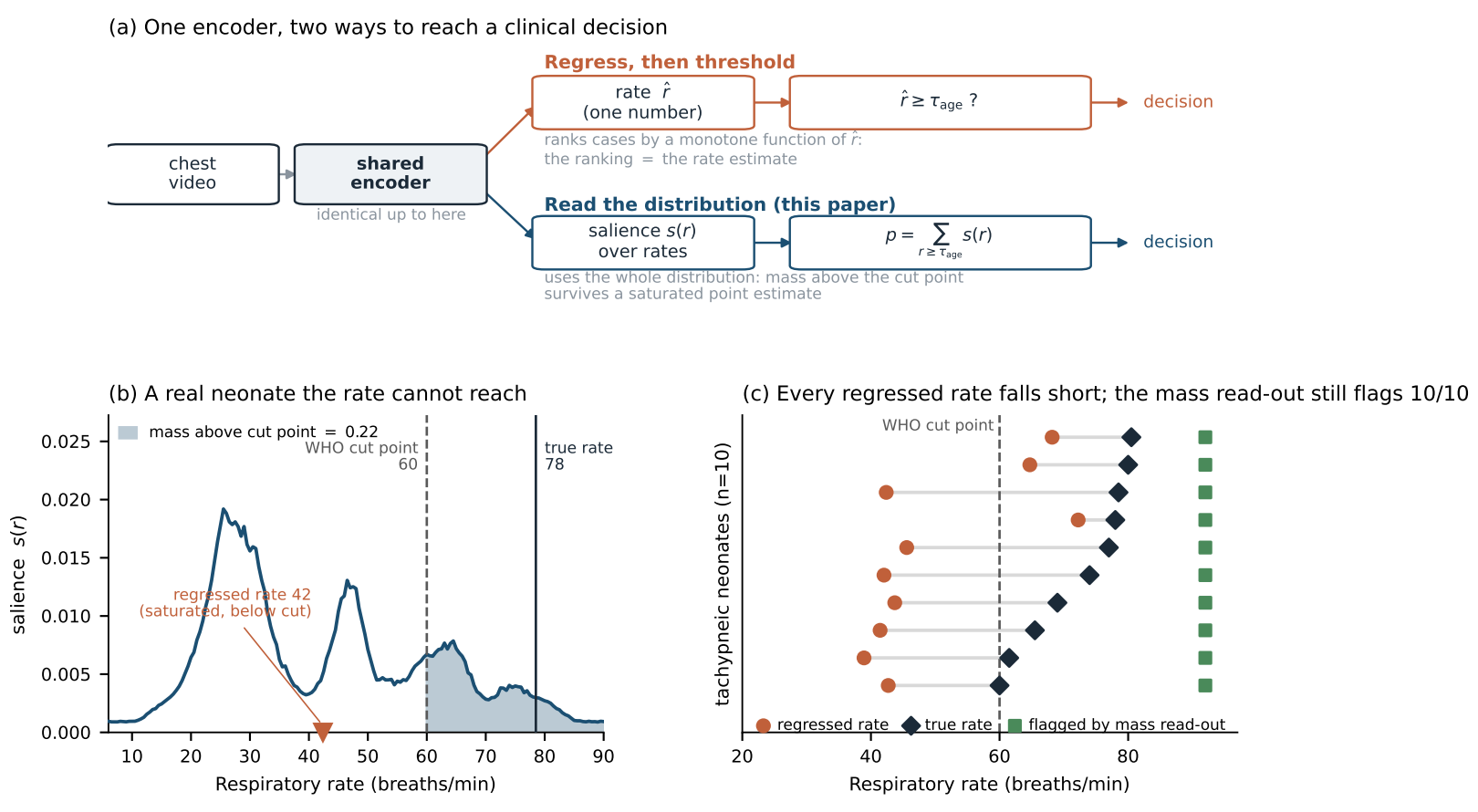
(a) One encoder, two ways to reach a clinical decision
Regress, then threshold
chest video
shared encoder
identical up to here
rate $\hat{r}$ (one number)
$\hat{r} \geq \tau_{age}$?
decision
ranks cases by a monotone function of $\hat{r}$:
the ranking = the rate estimate
Read the distribution (this paper)
salience $s(r)$ over rates
$p = \sum_{r \geq \tau_{age}} s(r)$
decision
uses the whole distribution: mass above the cut point
survives a saturated point estimate
(b) A real neonate the rate cannot reach
mass above cut point = 0.22
WHO cut point 60
true rate 78
regressed rate 42 (saturated, below cut)
salience $s(r)$
Respiratory rate (breaths/min)
(c) Every regressed rate falls short; the mass read-out still flags 10/10
WHO cut point
tachypneic neonates (n=10)
regressed rate
true rate
flagged by mass read-out
Respiratory rate (breaths/min)